NOAH FONG NEUWEG

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SUMMARY

Applied Math and Data Science student at UC San Diego who builds production data pipelines in Python and SQL. Designed and deployed a document extraction system at Scripps Institution of Oceanography that processed a 600-document corpus and surfaced 10 to 17 percent more records than manual expert review, at roughly eight cents per document. Two years of IT operations experience at UCSD's engineering school. U.S. citizen.

TECHNICAL EXPERIENCE

Automation and Data Engineering Intern
2026

Scripps Institution of Oceanography, UC San Diego — *San Diego, CA*

- Built an automated extraction pipeline for a 600-document corpus of cruise reports averaging 200 pages each, replacing a manual review process requiring roughly 8 hours and domain expertise per document.
- Validated output against manual extraction on 10 documents: captured every record identified by human reviewers plus 10 to 17 percent additional records missed in long tables, each verified against the source document.
- Reduced processing cost to approximately $0.08 per document through targeted prompt design and pre-extraction parsing with pdfplumber and PyMuPDF, constraining prompts to require source grounding for every extracted field.
- Delivered two deployments: a server version using FastAPI and uvicorn over PostgreSQL, orchestrated in n8n and containerized with Podman; and a fully local version running SQLite in a Docker container for environments where data cannot leave the premises.
- Designed the database schema and three workflows: ingestion, retrieval as structured JSON, and a reviewer interface allowing users to correct extractions and write changes back to the database. Built, documented, and handed off the full system independently, including GUI and maintenance documentation.
- Implemented a crosswalk standardization layer that learns canonical forms for entity and role names, normalizing inconsistent terminology across reports from different authors over time.

Student Computing Support Technician
2023 - Present

Jacobs School of Engineering, UC San Diego — *San Diego, CA*

- Image and deploy 30 or more systems per week, including OS installation, driver setup, and required software configuration.
- Perform secure data sanitization on hundreds of machines using 3-pass overwrite procedures, then prepare devices for reassignment.
- Diagnose and repair hardware and software issues across laptops and desktops, and maintain inventory and service logs supporting auditing and lifecycle tracking.

PROJECTS

Treasury Auction Scraper and Rate Tracker
2024 - Present

- Built a Python scraper for Treasury auction results using Selenium for JavaScript-rendered pages and BeautifulSoup for HTML parsing.
- Parsed and standardized auction tables into structured datasets, exporting cleaned CSV outputs and time-series plots comparing rate trends across maturities. Designed for scheduled runs and threshold-based alerting.

Match Outcome Classifier
2023 - 2025

- Processed 337,000 professional match records spanning 160 attributes across multiple seasons.
- Built an end-to-end ML workflow in Python covering preprocessing, feature engineering, train/test split, and evaluation.
- Trained a Random Forest classifier evaluated with accuracy and confusion-matrix diagnostics; identified time-based splits and calibration as next steps to reduce leakage risk.

SKILLS

Languages	Python, SQL, Bash
Data and ML	pandas, NumPy, scikit-learn, matplotlib, feature engineering, classification, regression, cross-validation, precision and recall, AUC
Backend	FastAPI, uvicorn, REST API design
Databases	PostgreSQL, SQLite, MySQL, schema design, joins and aggregation
Infrastructure	Docker, Podman, Linux, Git
Documents	pdfplumber, PyMuPDF, OCR workflows, LLM API integration with source-grounded prompting
Automation	n8n workflow orchestration, scheduled jobs, error handling, logging, Selenium, BeautifulSoup, Jupyter, Excel reporting

EDUCATION

B.S. Math and Applied Data Science, University of California, San Diego
Expected Dec 2026

Machine Learning Specialization, DeepLearning.AI and Stanford Online (Coursera) — Feb 2026

ACTIVITIES

PADI Open Water Scuba certified. Sailing (Quests and Sabots), rock climbing, running, and hiking.